

Problem

The voltage across a load and the current through it are given by

$$v(t) = 20 + 60 \cos 100t \text{ V}$$

$$i(t) = 1 - 0.5 \sin 100t \text{ A}$$

Find:

(a) the rms values of the voltage and of the current

(b) the average power dissipated in the load

Step-by-step solution

Step 1 of 2

(a)

$$V_{\text{rms}}^2 = 20^2 + \frac{60^2}{2}$$

$$V_{\text{rms}}^2 = 2200$$

$$V_{\text{rms}} = 46.9 \text{ V}$$

$$I_{\text{rms}} = \sqrt{1^2 + \frac{0.5^2}{2}}$$

$$I_{\text{rms}} = \sqrt{1.125}$$

$$I_{\text{rms}} = 1.061 \text{ A}$$

Step 2 of 2

(b)

$$P_{\text{avg}} = V_{\text{rms}} \times I_{\text{rms}}$$

$$P_{\text{avg}} = 46.9 \times 1.061$$

$$\therefore P_{\text{avg}} = 49.76 \text{ W}$$